Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

In which of the following pairs of compounds, nitrogen has maximum positive and maximum negative values of oxidation number?

(A) NH_3 and HN_3

(B) HNO_3 and HCN

(C) N_2O and NH_2OH

(D) N_2O_4 and NO_2

Q.2

Which of the following reactions is an example of redox reaction?

(A) $Cr_2O_3 + 6HCl \rightarrow 2CrCl_3 + 3H_2O$

(B) $2CrO_4^{2-} + H^+ \rightleftharpoons Cr_2O_7^{2-} + OH^-$

(C) $CrO_4^{2-} + 6I^- + 14H^+ \rightarrow 2Cr^{3+} + 3I_2 + 7H_2O$

(D) $CrO_3 + 2NaOH \rightarrow Na_2CrO_4 + H_2O$

Q.3

In the reaction

$C_2H_5OH + xI_2 + 6OH^- \rightarrow CHI_3 + HCO_2^- + yI^- + 5H_2O$

the value of x and y, respectively, are

(A) $2, 1$

(B) $3, 3$

(C) $4, 5$

(D) $5, 7$

Q.4

The stoichiometric numbers appearing from left to right in the reaction

$MnO_4^{2-} + H^+ \rightarrow MnO_4^- + MnO_2 + H_2O$

are

(A) $3, 2, 2, 1, 2$

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(B) $3, 4, 2, 1, 2$

(C) $2, 4, 1, 1, 2$

(D) $2, 3, 1, 1, 2$

Q.5

The increasing order of oxidation number of sulphur in the given compounds is

(A) $S_8 < S_2O_3^{2-} < S_4O_6^{2-} < S_2O_8^{2-}$

(B) $S_8 < S_2O_3^{2-} < S_4O_8^{2-} < S_4O_6^{2-}$

(C) $S_8 < S_2O_8^{2-} < S_2O_3^{2-} < S_4O_6^{2-}$

(D) $S_8 < S_2O_8^{2-} < S_4O_6^{2-} < S_2O_3^{2-}$

Q.6

Equivalent mass of $Na_2\ S_2O_3 \cdot 5H_2O$ in the reaction $2Na_2\ S_2O_3 + I_2 \rightarrow Na_2\ S_4O_6 + 2NaI$ is

(A) $124\ geq^{-1}$

(B) $248\ geq^{-1}$

(C) $84\ geq^{-1}$

(D) $72\ geq^{-1}$

Q.7

The oxidation number(s) of sulphur in pyrosulphurous acid $(H_2\ S_2O_5)$ is/are

(A) $+6, +6$

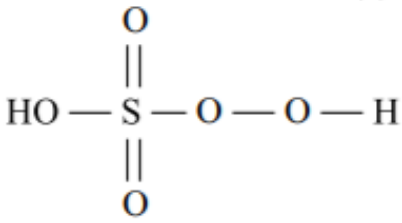
(B) $+5, +5$

(C) $+5, +3$

(D) $+4, +2$

Q.8

The peroxomonosulphuric acid is



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The oxidation number of S in this acid is

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(A) $+2$

(B) $+4$

(C) $+5$

(D) $+6$

Q.9

Which of the following is displacement reaction?

(A) $\mathrm{N_2(g) + O_2(\ g) \longrightarrow 2NO(g)}$

(B) $\mathrm{2Pb\,(NO_3)_2\,(s) \longrightarrow 2PbO(s) + 2NO_2(g) + \frac{1}{2}O_2(g)}$

(C) $\mathrm{NaH(s)\ H_2O(l) \longrightarrow NaOH(aq) + H_2(g)}$

(D) $\mathrm{2NO_2(g) + 2OH^{\ominus}(aq) \longrightarrow NO_2^{\ominus}(aq) + NO_3^{\ominus}(aq) + H_2O(l)}$

Q.10

Which of the following species do not show disproportionation reaction?

(A) $\mathrm{BrO^{\ominus}}$ (hypobromite ion)

(B) $\mathrm{BrO_2^{\ominus}}$ (bromite ion)

(C) $\mathrm{BrO_3{}^{\ominus}}$ (bromate ion)

(D) $\mathrm{BrO_4{}^{\ominus}}$ (perbromate ion)

Q.11

For the reaction

$\mathrm{3Br_2 + 6OH^{\ominus} \longrightarrow 5Br^{\ominus} + BrO_3^{\ominus} + 3H_2O}$

Equivalent weight of $\mathrm{Br_2}$ (molecular weight M) is

(A) $\frac{M}{2}$

(B) $\frac{M}{10}$

(C) $\left(\frac{M}{2}+\frac{M}{10}\right)$

(D) $\left(\frac{M}{6}\right)$

Q.12

$\mathrm{Cu_2S + MnO_4^{\ominus} \longrightarrow Cu^{2+} + Mn^{2+} + SO_2}$

The equivalent weight of $\mathrm{Cu_2}$ is

(A) $\frac{M}{2}$

(B) $\frac{M}{6}$

(C) $\frac{M}{8}$

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(D) $\frac{M}{4}$

Q.13

The equivalent weight of H_2SO_4 in the following reaction is

$Na_2Cr_2O_7 + 3SO_2 + H_2SO_4 \longrightarrow 3Na_2SO_4 + Cr_2(SO_4)_3 + H_2O$

(A) 98

(B) $\frac{98}{6}$

(C) $\frac{98}{2}$

(D) $\frac{98}{8}$

Q.14

In which of the following reactions, H_2SO_4 acts as an oxidising agent?

(A) $C_6H_{12}O_6 + H_2SO_4(conc) \longrightarrow 6C + 6H_2O$

(B) $5H_2SO_4(conc) + 4Zn \longrightarrow H_2S + 4Zn^{2+} + 4{SO_4}^{2-} + 4H_2O$

(C) $H_2SO_4(dil) + Zn \longrightarrow Zn^{2+} + H_2 + SO_4^{2-}$

(D) Both Option 2 and 3 are correct

Q.15

Oxidation number of S involved in coordination bond in $Na_2\ S_2O_3$ is

(A) -2

(B) $+2$

(C) $+5$

(D) -1

Q.16

How many of the following are examples of disproportionation reactions?

a. $Ag(NH_3)_2^{\oplus} + 2H^{\oplus} \longrightarrow Ag^{\oplus} + 2NH_4^{\oplus}$

b. $Cl_2 + 2\overset{\ominus}{O}H \longrightarrow ClO^{\ominus} + Cl^{\ominus} + H_2O$

c. $CaCO_3 \longrightarrow CaO + CO_2$

d. $2HgO \longrightarrow 2Hg + O_2$

e. $Cu_2O + 2H^{\oplus} \longrightarrow Cu + Cu^{2+} + H_2O$

f. $CuS + O_2 \longrightarrow Cu + SO_2$

g. $2HCuCl_2 +$ dilute with $H_2O \longrightarrow Cu + Cu^{2+} + 4Cl^{\ominus} + 2H^{\oplus}$

h. $2H_2O_2 \longrightarrow 2H_2O + O_2$

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Q.17

Balance the following by ion electron method in acidic medium.

$ClO_3^{\ominus} + I_2 \longrightarrow IO_3^{\ominus} + Cl^{\ominus}$

How many moles of H^+ are produced?

Q.18

$As_2S_3 + 7NaClO_3 + 12NaOH \longrightarrow 2Na_3AsO_4 + 7NaClO + 3Na_2SO_4 + 6H_2O$

The n factor of $As_2\ S_3$ is

Q.19

Balance the following reaction by ion electron method (acidic medium):

$As_2\ S_3 + NO_3^{\ominus} \longrightarrow S + NO_2 + AsO_4^{3-}$

How many moles of NO_3^- are required for the reaction?

Q.20

Balance the following reaction in acidic medium.

$CuS + NO_3^{\ominus} \longrightarrow Cu^{2+} + S_8 + NO$

How many moles of H_2O are produced?

Answers & Solutions

Q.1 Answer:

HNO_3 and HCN

Solution:

Nitrogen in HNO_3 and HCN has $+5$ and -3 oxidation number, respectively. These are the maximum positive or negative amongst the oxidation of number of N in other substances.

Q.2 Answer:

$CrO_4^{2-} + 6I^- + 14H^+ \rightarrow 2Cr^{3+} + 3I_2 + 7H_2O$

Solution:

In reactions (a), (b) and (d), no change in oxidation numbers of elements occur. Only in the choice (c), the oxidation number of Cr changes from $+6$ to $+3$ while that of I from -1 to 0 .

Q.3 Answer:

$4, 5$

Solution:

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The balancing of negative charges on both sides of the reaction suggests that the stoichiometric numbers of I_2 and I^- must be 4 and 5, respectively.

Q.4 Answer:

$3, 4, 2, 1, 2$

Solution:

The balanced equation is

$$3MnO_4^{2-} + 4H^+ \rightarrow 2MNO_4^- + MnO_2 + 2H_2O$$

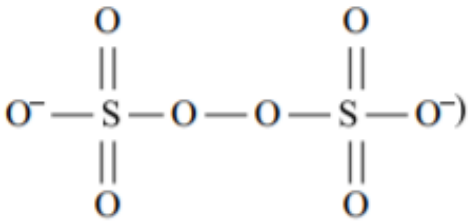
Q.5 Answer:

$S_8 < S_2O_3^{2-} < S_4O_6^{2-} < S_2O_8^{2-}$

Solution:

The oxidation number of S in the given compounds are 0 (for S_8), $+2$ (for $S_2O_3^{2-}$), $+2.5$ (for $S_4O_6^{2-}$) and $+6$

(for $S_2O_8^{2-}$ whose structure is



Q.6 Answer:

$248\ geq^{-1}$

Solution:

$2S_2O_3^{2-} \rightarrow S_4O_6^{2-} + 2e^-$

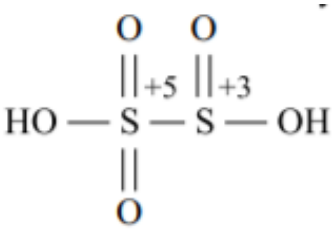
Equivalent mass of $Na_2S_2O_3 \cdot 5H_2O = \frac{2 \times \text{Molar mass of } Na_2S_2O_3 \cdot 5H_2O}{2 \text{ eq mol}^{-1}} = 248 \text{ g eq}^{-1}$

Q.7 Answer:

$+5, +3$

Solution:

The structure of $H_2S_2O_5$ is



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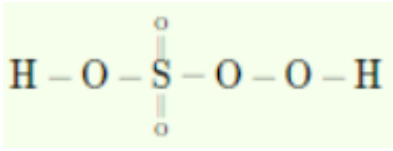
The oxidation number of sulphur atoms are $+5$ and $+3$, respectively.

Q.8 Answer:

$+6$

Solution:

H_2SO_5 has the following structure:



the oxygens in peroxide linkage i.e. $O-O$ has oxidation number $=-1$ each

and in oxide linkage i.e $=O$ and $H-O-S$ it is -2

Let the oxidation number of sulphur be x.

Oxidation number of $H=+1$

In $H_2SO_5: 2\times 1+x+3\times(-2)+2\times(-1)=0\, x+2-6-2=0$ or $x=+6$

thus Oxidation number of sulphur in peroxomonosulphuric acid (H_2SO_5) is $+6$.

Q.9 Answer:

$NaH(s)\ H_2O(l)\longrightarrow NaOH(aq)+H_2(g)$

Solution:

a. Combination redox reaction: N_2 and O_2 combines together to give NO (nitric oxide).

b. Decomposition redox (nitric oxide): $Pb(NO_3)_2$ (lead nitrate) breaks into three components.

c. Displacement redox reaction: Hydrogen of H_2O is displaced by $H^{\ominus}$ (hydride ion) into H_2 gas.

d. Disproportionation redox reaction: NO_2 ($+4$ oxida-tion state) disproportionates into $NO_2^{\ominus}$ ($+3$ oxidation state) and $NO_2^{\ominus}$ ($+5$ oxidation state).

So, the correct option is (c)

Q.10 Answer:

$BrO_4^{\ominus}$ (perbromate ion)

Solution:

$BrO_4^{\ominus}$ does not disproportionate because in this oxoanion,

Br is present in its highest oxidation state, i.e., $+7$. The disproportionation for other three oxoanions of Br is as follows:

$$3\,\overset{+1}{Br}\,O^{\ominus}\longrightarrow 2\,\overset{-1}{Br}^{\ominus}+\overset{+5}{Br}\,O_3^{\ominus}$$

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$$6\overset{+3}{\mathrm{Br}}\mathrm{O}_2^{\ominus} \xrightarrow{h\nu} 4\overset{+5}{\mathrm{Br}}\mathrm{O}_3^{\ominus} + 2\overset{-1}{\mathrm{Br}}^{\ominus}$$
$$4\overset{+5}{\mathrm{Br}}\mathrm{O}_3^{\ominus} \longrightarrow 3\overset{+7}{\mathrm{Br}}\mathrm{O}_4^{\ominus} + \overset{-1}{\mathrm{Br}}^{\ominus}$$

Q.11 Answer:

$\left(\frac{M}{2}+\frac{M}{10}\right)$

Solution:

Br_2 disproportionates (simultaneous oxidation and reduction), its equivalent weight is the sum of equivalent weights of the two half reactions.

$2e^- + \mathrm{Br}_2 \longrightarrow 2\mathrm{Br}^{\ominus} (x=2)$ **(reduction)**

$\mathrm{Br}_2 \longrightarrow 2\mathrm{BrO}_3^{\ominus} + 10\mathrm{e}^- \ (' n' = 10)$ **(oxidation)**

$2x = 0 \qquad 2x - 12 = -2$

$\qquad\qquad 2x = 10$

$\therefore Ew = \frac{M}{2} + \frac{M}{10} = \left(\frac{80\times 2}{2} + \frac{80\times 2}{10}\right) = 96$

Q.12 Answer:

$\frac{M}{8}$

Solution:

Both Cu^{1+} and S^{2-} in $\mathrm{Cu}_2\,\mathrm{S}$ undergo oxidation.

$\mathrm{Cu}_2^{+1\times 2} \longrightarrow 2\mathrm{Cu}^{2+} + 2e^- \ (' n' = 2)$ **(oxidation)**

$2x = 2 \qquad 2x = 4$

$\mathrm{S}^{2-} \longrightarrow \mathrm{SO}_2 + 6e^- \ (' n' = 6)$ **(oxidation)**

$x = -2 \qquad x - 4 = 0$

$\qquad\qquad x = 4$

Total $e^- = 8$

$Ew = \frac{M}{8}$

Q.13 Answer:

$\frac{98}{2}$

Solution:

In this reaction, 6 mol of e^- are involved with 1 mol of $\mathrm{H_2SO_4}$ in this redox reaction. $\mathrm{H_2SO_4}$ acts here as acidic medium $(' n' = 2)$, $(2\mathrm{H}^{\oplus})$.

So, Ew of $\mathrm{H_2SO_4} = \frac{M}{2} = \frac{98}{2} = 49$

Q.14 Answer:

Both Option 2 and 3 are correct

Solution:

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a. As a dehydrating agent, since it simply removes water from glucose.

b. $H_2SO_4(\text{ S in } +6 \text{ state }) \longrightarrow H_2S$ **(S in** -2 **state), i.e., acid is reduced. So it acts as an oxidising agent.**

c. All strong acids liberate H_2 **with active metals such as** Zn **(i.e., it is reduced) unless** ${SO_4}^{2-}$ **is reduced. In this case,** $H^{\oplus}$ **is reduced to** H_2**. So it acts as an oxidising agent. We can also call it as an acidic property.**

Q.15 Answer:

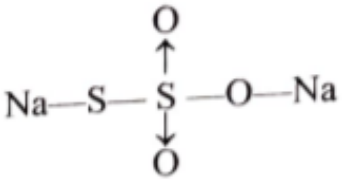
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Solution:

$Na_2\mathbf{S_2}O_3 : 2 \times 1 + 2 \times a + 3(-2) = 0 \Rightarrow a = +2$

Here too, it is the average oxidation number.

The structure of $Na_2S_2O_3$ **is**



The oxidation number of S **involved in coordinate bond,**

i.e., donor S **atom, is** $+5$**. The oxidation number of other** S **atom is** -1**.**

Q.16 Answer:

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Solution:

A reaction in which same atom is oxidised and reduced at the same time is called as a disproportionation reaction. (b), (e), (g), and (h) are disproportionation reactions.

Q.17 Answer:

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Solution:

Two half reactions are

a. $ClO_3^{\ominus} \longrightarrow Cl^{\ominus}$

Balancing them separately:

$ClO_3^{\ominus} \longrightarrow Cl^{\ominus}$

$ClO_3^{\ominus} + 6H^{\oplus} \longrightarrow Cl^{\ominus} + 3H_2O$

$ClO_3^{\ominus} + 6e^- + 6H^{\oplus} \longrightarrow Cl^{\ominus} + 3H_2O\big] \times 5$

b. $I_2 \longrightarrow IO_3^{\ominus}$

$I_2 + 6H_2O \longrightarrow 2IO_3^{\ominus} + 12H^{\oplus}$

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$I_2 + 6H_2O \longrightarrow 2IO_3^{\ominus} + 10e^- + 12H^{\oplus}\big] \times 3$

Balance electrons in two half reactions, and adding we get:

$5ClO_3^{\ominus} + 3I_2 + 3H_2O \longrightarrow 5Cl^{\ominus} + 6IO_3^{\ominus} + 6H^{\oplus}$

Q.18 Answer:

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Solution:

Both ${As_2}^{+3\times2}$ **and** $S_3^{-2\times3}$ **undergo oxidation.**

${As_2}^{+3\times2} \longrightarrow 2AsO_4^{3-} + 4e^-$ **(oxidation)**

$2x = +6 \qquad 2x - 16 = -6$

$2x = 10$

$S_3^{-2\times3} \longrightarrow 3SO_4^{2-} + 24e^- \quad$ **(oxidation)**

$3x = -6 \qquad 3x - 24 = -6$

$3x = 18$

Total $e^- = 24 + 4 = 28$

$Ew = \frac{M}{28}$

Q.19 Answer:

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Solution:

This is a very special case of redox reaction.

Note that the given reaction involves two oxidations and one reduction.

a. Two oxidation half reactions are

i. $As^{3+} \longrightarrow AsO_4^{3-}$

ii. $S^{2-} \longrightarrow S$

First balancing these two half reductions:

i. $2As^{3+} + 8H_2O \longrightarrow 2AsO_4^{3-} + 16H^{\oplus} + 4e^-$

ii. $3S^{2-} \longrightarrow 3\,S + 6e^-$

Now add the two equations to get

$2As^{3+} + 3S^{2-} + 8H_2O \longrightarrow 2AsO_4^{3-} + 3S + 16H^{\oplus} + 10e^- \ldots (i)$

b. Reduction half reaction: $NO_3^{\ominus} \longrightarrow NO_2$

Now balance the reduction half reaction and proceed as usual.

$NO_3^{\ominus} + 2H^{\oplus} + 1e^- \longrightarrow NO_2 + H_2O \ldots (ii)$

Adding the two reactions (i) **and** (ii)**, we get**

$As_2S_3 + 10NO_3^{\ominus} + 4H^{\oplus} \longrightarrow 2AsO_4^{3-} + 10NO_2 + 3S + 2H_2O$

Note: In the above type of cases, take total change in oxidation number (increase or decrease) for cross multiplication while adding oxidation and reduction half reactions. For example, in the given case, 10 **electrons are lost per mole of** As_2S_3**.**

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Q.20 Answer:

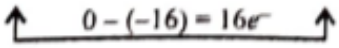
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Solution:

(Ion electron method)

i. $8S^{2-} \longrightarrow S_8 + 16e^-$ **(Oxidation)**

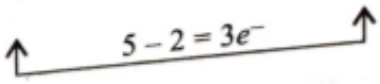
$8x = -16 \qquad 8x = 0$



ii. $3e^- + NO_3^{\ominus} \longrightarrow$ **NO** **(Reduction)**

$x - 6 = -1 \qquad x - 2 = 0$

$x = 5 \quad x = 2$



iii. Balance O **atoms in Eq. (ii) by adding** $2H_2O_{to}$ RHS

$3e^- + NO_3^{\ominus} \longrightarrow NO + 2H_2O$

iv. Balance H **atoms in Eq. (iii) by adding** $4H^{\oplus}$ **ions to** LHS

$4H^{\oplus} + 3e^- + NO_3^{\ominus} \longrightarrow NO + 2H_2O$

v. Multiply Eq. (i) by 3 **and Eq. (iv) by** 16 **and add them**

$$24S^{2-} \longrightarrow 3S_8 + 48e^-$$

$64H^{\oplus} + 48e^- + 16NO_3^{\ominus} \longrightarrow 16NO + 32H_2O$

$24S^{2-} + 64H^{\oplus} + 16NO_3{}^{\ominus} \longrightarrow 3\ S_8 + 16NO + 32H_20$

Add $24Cu^{2+}$ **to both sides.**

vi. $24CuS + 64H^{\oplus} + 16NO_3^{\ominus} \longrightarrow 24Cu^{2+} + 3\ S_8 + 16NO + 32H_2O$

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